

SEVERNOYE, Russian Federation

WMO: 294180

Lat: 56.3377N

Lon: 78.3575E

Elev: 415

StdP: 14.48

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.7	-27.8	-38.6	0.6	-33.4	-33.4	0.8	-27.2	22.3	11.5	20.0	18.9	1.4	350	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	83.3	66.0	80.4	65.1	77.7	63.7	69.1	78.7	67.4	76.6	65.7	74.2	7.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.7	96.4	73.8	64.0	90.7	71.4	62.4	85.6	70.2	33.6	78.7	32.2	76.6	30.9	74.2	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	17.8	15.3	-40.9	87.5	7.2	2.7	-46.1	89.5	-50.3	91.1	-54.4	92.6	-59.6	94.6		
(5)			-40.8	72.4	6.8	1.6	-45.7	73.5	-49.7	74.5	-53.6	75.4	-58.5	76.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.6	-2.1	3.2	19.7	36.5	51.3	61.3	64.7	59.4	48.7	36.2	17.9	4.5		
	DBStd	25.60	14.81	13.25	10.73	10.61	9.73	7.76	6.01	6.57	7.83	8.98	14.25	15.36		
	HDD50	7326	1615	1310	939	418	106	10	0	5	116	436	962	1409		
	HDD65	11603	2080	1730	1404	854	433	157	80	195	490	894	1412	1874		
	CDD50	1346	0	0	0	14	147	348	456	297	77	7	0	0		
	CDD65	149	0	0	0	0	9	45	71	22	2	0	0	0		
	CDH74	1521	0	0	0	2	177	473	604	240	25	0	0	0		
	CDH80	284	0	0	0	0	33	96	117	36	2	0	0	0		
	WSAvg	6.3	6.2	6.8	7.7	7.4	6.9	5.2	4.5	4.8	5.6	6.7	7.0	7.3		
	PrecAvg	18.7	0.9	0.7	0.7	1.0	1.5	2.5	3.1	2.7	1.7	1.5	1.4	1.1		
(15) Precipitation	PrecMax	25.8	1.8	1.4	2.0	2.7	3.3	5.9	8.9	5.2	3.8	3.0	2.7	2.2		
	PrecMin	12.5	0.1	0.0	0.0	0.3	0.4	0.4	0.4	0.5	0.5	0.6	0.4	0.3		
	PrecStd	3.8	0.5	0.3	0.5	0.6	0.7	1.4	1.8	1.2	0.9	0.7	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.5	33.7	45.2	71.0	84.3	86.4	87.1	84.0	77.1	64.9	44.0	34.2		
		MCWB	30.1	31.3	37.4	54.5	60.7	66.6	69.6	67.4	60.6	51.6	41.0	32.5		
	2%	DB	27.3	30.2	40.2	63.2	78.8	82.8	83.1	79.8	71.8	59.2	40.4	31.9		
		MCWB	26.0	28.7	34.3	50.1	59.0	65.3	67.1	65.7	57.9	48.8	38.2	30.5		
	5%	DB	24.0	26.9	37.1	57.7	74.5	79.5	80.3	76.5	66.6	53.4	36.9	28.7		
		MCWB	22.9	25.5	32.9	47.0	56.9	64.6	66.2	64.1	55.6	45.4	34.7	27.4		
	10%	DB	19.9	23.0	34.5	52.2	69.4	76.0	77.3	72.8	62.2	48.9	33.8	25.0		
		MCWB	18.9	21.6	31.3	43.7	54.8	62.9	65.3	62.4	53.3	43.6	32.2	24.0		
	0.4%	WB	30.4	31.8	37.8	55.6	63.6	71.0	71.9	70.0	62.9	53.2	41.7	33.1		
		MCDB	31.4	33.2	43.8	68.0	80.7	81.0	82.5	79.2	73.3	62.2	42.8	33.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.1	28.6	35.3	51.3	60.8	68.0	69.7	67.5	59.3	49.6	38.4	30.6		
		MCDB	27.2	30.1	38.8	60.5	75.4	78.4	79.4	77.5	68.3	57.8	40.3	31.9		
	5%	WB	22.8	25.5	33.5	47.7	58.0	66.1	68.2	65.6	56.9	46.7	35.0	27.5		
		MCDB	23.9	26.8	36.6	56.5	71.5	76.4	76.9	73.4	65.0	52.1	36.6	28.6		
	10%	WB	18.9	21.6	31.5	44.5	55.5	64.2	66.5	63.5	54.3	43.8	32.5	24.0		
		MCDB	19.9	23.0	34.5	51.9	67.9	73.5	74.4	70.8	61.2	48.5	33.7	25.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	17.6	17.4	17.0	20.4	19.5	18.3	18.6	17.3	13.0	11.3	14.0		
		MCDBR	14.5	13.5	16.8	24.5	27.8	25.0	23.0	25.1	25.1	22.0	10.7	13.0		
		MCWBR	14.2	12.7	12.6	14.0	12.2	10.9	9.8	12.6	13.3	13.1	9.7	12.8		
		MCDBR	14.4	13.5	13.9	22.9	25.6	21.7	20.0	21.5	22.3	20.2	10.2	13.2		
	5% WB	MCWBR	14.2	13.1	11.2	13.8	12.9	10.7	9.7	12.0	12.7	13.0	9.5	13.0		
(40) Clear-Sky Solar Irradiance	taub		0.280	0.313	0.368	0.406	0.402	0.404	0.434	0.412	0.364	0.349	0.310	0.283		
	taud		2.023	2.004	1.927	2.069	2.235	2.301	2.213	2.252	2.347	2.234	2.077	1.966		
	Ebn at Noon		182	226	241	254	265	266	254	252	249	214	168	142		
	Edh at Noon		20	31	44	45	40	38	41	37	29	25	19	17		
(44) All-Sky Solar Radiation	RadAvg		260	587	1075	1460	1657	1849	1672	1323	912	454	216	161		
	RadStd		22	36	56	120	148	190	115	131	94	44	41	22		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page